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UCSD Resume - Winter 2026.docx

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Jonathan Longhurst Maku

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EDUCATION

University of California, San Diego - B.S. Chemical Engineering *Expected Graduation: Spring 2027*

Minor: Visual Arts: Photography

Specializations: Process Dynamics and Control & Thermal Engineering, and Systems.

Relevant Coursework: Fluid Dynamics, Thermodynamics, Nuclear Energy, Electrochemistry, Analog Design, Circuitry.

EXPERIENCE

Qualcomm Institute, San Diego, CA

September 2025 - Present

Equipment Engineering Intern, Nano3 Cleanroom Facility.

- Performing maintenance, repair, and rebuilding of advanced laboratory equipment used for nano & bioengineering research.
- Guiding and supporting industry and academic professionals through training in equipment use, and **laboratory safety**.
- Managed facility standards by performing bead blasting for system decontamination & ensuring chemical safety compliance.

Qualcomm Institute, San Diego, CA

December 2025

May 2025 -

Laboratory Assistant, Nano3 Cleanroom Facility.

- Supporting day-to-day lab operations by managing inventory, restocking chemicals, and maintaining an organized lab space.
- Assisting with hazardous waste management by organizing chemical waste in compliance with **safety protocols**.
- Ensuring adequate **lab safety** levels and upholding cleanliness standards to support hazard-free research environments.

Envision Engineering & Arts Makerspace (UCSD), San Diego, CA

Engineering Lab

Assistant

April 2025

- September 2025

- Provided guidance to students on **fabrication and design equipment**, including 3D printers, laser cutters, CNC milling machines, power tools, table saws, and drill presses, ensuring adherence to safety protocols and optimal equipment utilization.
- Delivered hands-on workshop sessions on topics such as analog prototyping, **CAD modeling**, and interactive electronics.
- Immersed myself in an engineering space, developing my knowledge on students' projects and solutions to common issues.

Recreation (UCSD), San Diego, CA

Facility Supervisor

March 2024 -

May 2025

- **AWARDS:** *Employee of the Month:* November 2024, *Rookie of the Year Finalist:* May 2025.
- Oversaw a team of **100+** student employees across three fitness facilities, which serve **5000+** students daily.
- Managed daily facility operations, including opening/closing procedures, equipment checks, and cleanliness standards.

Student Affairs (UCSD), San Diego, CA

January 2024 -

June 2024

Events Intern

- Planned, coordinated, and executed programming and events, which **1500+** students attended regularly.
- Managed **\$2000-\$5000** quarterly, tracking on **Excel spreadsheets**, to budget programming efforts in UCSD spaces.

PROJECTS

Chemically Powered Miniature Vehicle - AIChE, Mechanical Lead Position.

September

2024 - Present

- Manufacturing and constructing a miniature vehicle powered by a Lead-Acid **electrochemical reaction**.
- Simulating and modeling the vehicle's mechanics, casing, and components using **SolidWorks** and **Autodesk Fusion**.
- Collaborating in a team of **40+** people and leading my sub-team of **10** to create the mechanical components of the vehicle.

Hydrogen Fuel Cell - AIChE, Process Engineer Position.

May 2025 - Present

- Engineering a portable **hydrogen fuel cell** system aimed at powering a **sustainable energy** conversion system.
- **Designing** and **3D printing** custom parts and enclosures using **Autodesk Fusion** to support gas containment and

flow control.

- Collaborating in a multidisciplinary team to integrate chemical, mechanical, and electrical teams for optimal performance.

Rocket - Rocket Propulsion Lab, Project Engineer

Position.

September 2024 - May

2025

- Conceived, constructed, launched, and recovered a **24-30 inch** solid propellant motor-powered rocket. **3000+ ft altitude.**
- Designed, simulated, and modeled a model rocket using **SolidWorks, OpenRocket, and MATLAB** software.
- Programmed avionics with **Arduino** and **C++** programming software, connected with protoboards and breadboards.
- Manufactured and assembled utilizing **3D printing, soldering, and power tools:** table saw, laser cutting, etc.

SKILLS

Languages: Fluent in Spanish, Intermediate in German.

Technical Skills: C++, Python, SolidWorks, AutoCAD, MATLAB, OpenRocket, Excel, Arduino, 3D Printing, Raspberry Pi.

Machinery Proficiency: Lab Safety, Chemical Experience, Laser cutting, CNC Machining, Table Saws, Drill Press, etc.

Soft Skills: Quick-Learner, Teamwork, Communication, Planning, Problem-Solving, Organization, Adaptivity, Creativity.

